

2(a)(i)	$pV = nRT$	C1
	$n = (3.0 \times 10^5 \times 210 \times 10^{-6}) / (8.31 \times 270)$	C1
	$= 0.028 \text{ mol}$	A1
2(a)(ii)	$V \propto T$ or $T = pV / nR$ with value of n from (i)	C1
	$T = (140 / 210) \times 270$ or $T = (3.0 \times 10^5 \times 140 \times 10^{-6}) / (8.31 \times 0.028)$	A1
	$= 180 \text{ K}$	
2(a)(iii)	$W = p\Delta V$ $= 3.0 \times 10^5 \times (210 - 140) \times 10^{-6}$	C1
	$= 21 \text{ J}$	A1

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Question	Answer	Marks
2(b)	$\Delta U = w + q$	C1
	$= 21 - 53$	C1
	or	
	$\Delta U = (nN_A) \times (3/2)k\Delta T$	(C1)
	$= (0.0281 \times 6.02 \times 10^{23}) \times (3/2) \times 1.38 \times 10^{-23} \times (180 - 270)$	(C1)
	or	
	$\Delta U = (3/2)nR\Delta T$	(C1)
	$= (3/2) \times 0.0281 \times 8.31 \times (180 - 270)$	(C1)
	$\Delta U = (-)32 \text{ J}$	A1